



City Climate
Finance Gap Fund



WORLD BANK GROUP

FINAL REPORT

BOGOTA LOW-CARBON VITAL NEIGHBORHOODS ASA Case Study: Bogota's Vital Neighborhoods Strategy

PROJECT P177851

Urban, Disaster Risk Management, Resilience, and Land Global Practice (GPURL) -
Latin America and the Caribbean

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ABBREVIATIONS AND ACRONYMS

BACA

Bogotá a Cielo Abierto (Bogotá Open Sky)

AE

Actuaciones Estratégicas (Strategic Actions)

AIM

Áreas de Integración Modal (Modal Integration Areas)

DANE

Departamento Nacional de Estadística (National Administrative Department of Statistics)

DEMOS

Distritos Especiales de Mejoramiento y Organización Sectorial (Special Districts for Improvement and Sectoral Organization)

EEP

Estructura Ecológica Principal (Main Ecological Structure)

EIA

Evaluación de Impacto Ambiental (Environmental Impact Assessment)

GHG

Gases Efecto Invernadero (Greenhouse Gas)

GDP

Producto interno bruto (Gross domestic product)

IDU

Instituto de Desarrollo Urbano (Urban Development Institute)

LCVN

Barrios vitales bajos en carbono (Low-carbon vital neighborhoods)

LVC

Instrumentos de Captura de Valor del Suelo (Land Value Capture)

MdC

Manzanas del Cuidado (Care Blocks)

NbS

Soluciones basadas en la naturaleza (Nature based solutions)

ND-GAIN

Iniciativa de Adaptación Global de Notre Dame (Notre Dame Global Adaptation Initiative)

OECD

Organization for Economic Co-operation and Development

PAC

Plan de Acción Climática (Climate Action Plan)

PIA

Evaluación de impacto participativa (Participatory Impact Assessment)

POT

Plan de Ordenamiento Territorial (Land Use Plan)

PRUM

Proyectos de Renovación Urbana para la Movilidad Sostenible (Urban Renewal Projects for Sustainable Mobility)

REA

Evaluación Rápida de Impacto Ambiental (Rapid Environmental Impact Assessment)

SDA

Secretaría Distrital de Ambiente (District Environment Secretariat)

SDCRD

Secretaría de Cultura, Recreación y Deporte (District Culture, Recreation and Sport Secretariat)

SDDE

Secretaría Distrital de Desarrollo Económico (District Economic Development Secretariat)

SDHT

Secretaría Distrital de Hábitat (District Habitat Secretariat)

SDM

Secretaría Distrital de Movilidad (District Mobility Secretariat)

SDMujer

Secretaría Distrital de la Mujer (District Woman's Secretariat)

SDP

Secretaría Distrital de Planeación (District Planning Secretariat)

SIA

Evaluación de impacto de sostenibilidad (Sustainability Impact Assessment)

SUDS

Sistemas urbanos de drenaje sostenible (Sustainable urban drainage systems)

UPL

Unidad de Planeamiento Local (Local Planning Unit)

UPZ

Unidad de Planeamiento Zonal (Zonal Planning Unit)

1. INTRODUCTION

1. Over the last half century, Colombian cities have experienced an accelerated process of urbanization and expansion of their urban footprint. The population living in cities rose from 46 percent in 1960 to 82 percent in 2021, and is expected to reach 88 percent by 2050 (World Urbanization Prospects, 2018). Likewise, between 1990 and 2015, cities expanded their urban footprints by an average of 2.5 percent per year, and the urban population grew by 2.3 percent annually (OECD. Colombia Urban Policy Review, 2022).

2. Cities are drivers of productivity, engines of development and vehicles for social and economic mobility. Colombia's urban areas generate close to 85 percent of the country's gross domestic product (GDP), with the five main cities (Barranquilla, Bogotá, Cali, Cartagena and Medellín) contributing 40 percent of the total. Even though these cities have demonstrated sustained economic growth over recent years, productivity levels remain among the lowest in OECD countries. Furthermore, their land use model (in low density and urban sprawl) is inadequate to respond to challenges such as: i) increasing vehicular congestion and air pollution; (ii) a rise in road accidents; (iii) high crime rate and perception of insecurity; (iv) limited access to public services; and (v) threats to ecosystems and lack of resilience to climate disasters.

3. The Covid-19 pandemic has exacerbated these challenges by reversing the downward trend in poverty and inequality and intensifying the country's climate emergency. According to the Great Integrated Household Survey (GEIH), unemployment went from 10.9 percent in 2019 to 16.5 percent in 2020 (DANE, 2022). Colombia also ranks 84th out of 181 countries for extreme weather events such as droughts and hurricanes, as per the Country Index of the Global Adaptation Initiative (ND-GAIN) of the University of Notre Dame. These events are expected to become more frequent and destructive as the temperature rises to an estimated 3.50 C by 2071. In response, the Government of Colombia has enacted the Climate Action Law of 2021, aiming for carbon neutrality by 2050. The Law mandates a 51

percent reduction in greenhouse gas (GHG) emissions by 2050, and the incorporation of climate change adaptation into territorial instruments by 2030.

4. Bogotá, as the main urban center and capital of the country, contributes 25.3 percent to the national GDP, the most of any urban area (DANE, 2021). However, because of unsustainable land use patterns, Bogotá's physical and natural environment is deteriorating. With its population amounting to 15 percent of the country's total (7,968,095 of 52,156,254 million), -(DNP, 2023), the capital attracts specialized industries and talent, provides opportunities for economies of agglomeration and scale, and enhances competitiveness and productivity. However, the city has increased its urban footprint by 15.34 percent between 1997-2016, and if this rate of growth continues, the urban footprint will exceed more than 50 percent of the city's territory by 2050.¹ Further, due to urban sprawl, the city has doubled the number of vacant houses since 2005 (118,753 vacant houses as of 2018), with the largest increases in the central areas; while at the same time, the central areas concentrate the highest percentage of dwellings that encroach into public spaces.

5. In response to these growth challenges, Bogotá has strengthened its urban planning, management and financing instruments to accomplish a model of proximity that is resilient to climate change. This approach hinges on *three key principles of urban design*: (i) *Proximity* focuses on ensuring that urban residents can access services and amenities within a 15 or 20-minute walk, without having to pay for travel or spend time in traffic;² (ii) *Care* refers to a set of services, regulations, policies, and technical and institutional actions that seeks to recognize, redistribute, and reduce care work, understanding it as a necessary social function for the daily life of people and the functioning of society; and (iii) *Resilience* is related to the capacity of social, economic, and environmental systems to cope with a dangerous event, trend, or disturbance to respond or reorganize themselves in a way that maintains their essential function, identity and structure, while at the same time preserving the capacity for adaptation, learning and transformation (Law 1931 of 2018).

¹ IDOM, 2019. Study of Growth and Evolution of the Urban Footprint for Bogotá Region. Retrieved from: https://www.sdp.gov.co/sites/default/files/7_estudio_de_huella_urbana_0.pdf

² Moreno, Carlos (2022). LIVING IN PROXIMITY IN A LIVING CITY. Glocalism: Journal of Culture, Politics and Innovation. He introduced concept of chrono-urbanism from which these principles of urban design are derived. Chrono-urbanism is defined as the maximum time that people are willing to walk to satisfy their daily needs in their locality, which in a city is estimated between 15 and 20 minutes and represents a walk of 800 meters from the house to the destination and return.

6. Between 2020 and 2022, Bogotá adopted four interrelated planning instruments through which these design principles will be implemented³. During this period, the city proposed a range of interventions to improve the accessibility of urban services. Of these proposed interventions, the Vital Neighborhoods (VN) Strategy has been prioritized in the Land Use Plan as a mechanism to catalyze urban transformation. This strategy was chosen for its potential to i) promote low carbon transport modes (transit, pedestrians, bicycles; ii) decrease traffic speed on neighborhood streets; (iii) foster citizens' engagement with public spaces; and (iv) stimulate economic development within beneficiary communities.

7. In addition, in line with the city's commitment to sustainable urban development⁴, the Land Use Plan also considered the implementation of 33 Low Carbon Vital Neighborhoods (LCVN), one for each Local Planning Unit (UPL).⁵ However, despite the efforts of the city administration to prioritize five pilot neighborhoods for implementation of the LCVN Strategy,⁶ have been hampered by the lack of a governance scheme that would ensure the sustainability of the strategy over time and provide a channel for funneling economic resources to those neighborhoods. Despite these limitations, the first pilot in the San Felipe neighborhood conducted from 2020 to 2022, led to significant improvements: a 27 percent reduction in the road accidents victims, a 20 percent decrease in thefts compared to 2021, and a substantial 168 boost in monthly retail sales due to the recovery of public space and economic revival of the neighborhood (SDM, 2022).

8. To support Bogotá's urban transformation, the World Bank, through the City Climate Finance Gap Fund (GAP FUND), has been providing technical assistance to the city administration. The technical assistance seeks to support the city in transforming its climate ambitions⁷ into projects ready for financing, and in facilitating access to tools and platforms for resilient urbanization and low carbon emissions at the regional, national and local levels. The

inputs provided by this technical assistance made it possible to identify that the path towards implementing Bogotá's Vital Neighborhoods Strategy faces three pivotal challenges: (i) the lack of a roadmap for inter-institutional convergence, which has limited the ability of the city's administration to activate mechanisms subject to new regulations or specified in the existing legal framework; (ii) integrating neighborhoods into local economic and transportation networks necessitates a nuanced understanding of the urban fabric; and (iii) to assess the environmental impact of urban transformations and climate change mitigation and adaptation efforts the establishment of a consistent monitoring and evaluation system is critical.

9. This report summarizes the activities, findings and lessons from the pilots carried out in the city, along with recommendations and next steps toward achieving the proximity model under the lens of resilience to climate change. The report consists of this Introduction, an Executive Summary (Section 2) and four additional sections. Section 3 analyzes the city's legal framework, which codifies the basic elements of the LCVN Strategy within the Land Use regulations and proposes a management model to facilitate implementation of the initiative. Section 4 establishes the guidelines and design parameters required to promote a city model under the principles of ecosystemic urbanism⁸ and sustainable mobility. It also proposes a spatial configuration for two of Bogotá's urban areas – the San Felipe and San Carlos neighborhoods – that have different types of economic and environmental assets. Section 5 explores the environmental benefits derived and that could result from urban design actions, suggests a baseline and measurement indicators, and outlines the roadmap to replicate these actions in other neighborhoods. Moreover, it also specifies the environmental guidelines regarding mitigation and adaptation actions for the San Carlos neighborhood. Finally, Section 6 presents the steps to be followed by the city to manage the implementation and scalability of the proximity model project.

³ The Land Use Plan (POT; Bogotá Reverdece 2022-2035), proposes a new model for a compact city. The Economic, Social, Environmental and Public Works Development Plan (A New Social and Environmental Contract for Bogotá in the 21st Century: 2020-2024) proposes strategies for sustainable and accessible mobility. The Master Plan for Safe and Sustainable Mobility (PMSS) based on proximity and chrono-urbanism. And the Climate Action Plan (PAC) 2020-2050 provides a vision for resilient adaptation to climate change in the medium and long term.

⁴ In accordance with Article 7 of the POT, this concept refers to the recognition of the local scale as the ideal territorial area for the planning and management of the Bogotá territory, as part of a commitment to justice and territorial balance, and in response to the challenges of a close and caring territory in the context of the pandemic.

⁵ The UPLs seek to guarantee the minimum conditions of proximity, availability and diversity of territorial supports, social and care services, and access to employment and improve commuting conditions, also facilitating integration, productivity and competitiveness. Each UPL will not exceed 400,000 inhabitants, and will be delimited respecting the social fabric and cultural identity of the territories, their economic dynamics, access to different essential services and equitable distribution of their inhabitants.

⁶ San Felipe – Juan XXII (Barrios Unidos), 20 de Julio–San Blas (San Cristóbal), El Porvenir (Bosa), Las Cruces– San Bernardo (Santa Fé) and San Carlos (Tunjuelito)

⁷ In accordance with Bogotá's PAC, by 2050 the city aims to confront the climate crisis and consolidate itself as a more resilient and sustainable city.

⁸ Ecosystemic urbanism considers the city as an ecosystem with the intention of projecting itself into the future and is a tool that allows solving the great challenges of cities. Its proposed city model seeks (i) to be compact in its morphology and to make things close; (ii) it is based on organizational complexity; (iii) it is related to urban metabolism, always seeking efficiency and the minimum impact on land systems, and (iv) it seeks to achieve maximum social cohesion.

2. EXECUTIVE SUMMARY

10. The city's administration requires laying the foundations for the effective and long-term implementation of the LCVN Strategy. To support the fulfillment of this purpose, this Report analyzed the city's institutional capacity for land use regulation and recommended that the administration must: (i) identify appropriate institutional and governance arrangements to implement the LCVN strategy; (ii) design a management model with viable financial and management mechanisms; and (iii) strengthen the San Felipe pilot and ensure that its model is scalable. The analysis also included recommendations for how to formulate the Sustainable and Safe Mobility Plan (PMSS) of Bogotá. This urban planning instrument, which complements the Land Use Plan, was put in place to guide mobility planning towards achieving proximity.

11. The unsustainable urban growth model of the city suggests the need to apply the bases of ecosystemic urbanism and sustainable mobility, for which design guides aligned with the proximity city model were developed. To achieve this goal, three strategic axes were defined following models such as the Superblocks in Barcelona, Low-Traffic Neighborhoods in London, and Ville du Quart d'Heure in Paris (Figures 1-3): (i) Reorganization of mobility favoring sustainable modes of transport; (ii) revitalization of public space and infrastructure, and (iii) strengthening of community services considering the basic needs of people. With the implementation of these design guidelines, an increase in green areas, public space, pedestrian circulation areas, more roads with speed limits and better circulation redistribution are expected. The San Felipe and San Carlos neighborhoods were selected as case studies.

Figure 1. Barcelona's Superblocks



Figure 2. London's Low-Traffic Neighborhoods



Figure 3. Paris' Ville du Quart d'Heure



Sources:

Figure No 1. <https://www.barcelona.cat/pla-superilla-barcelona/es#fancybox=1049-3>

Figure No 2. <https://www.theguardian.com/environment/2020/nov/20/so-you-want-to-set-up-a-low-traffic-neighbourhood-heres-why-re-to-start>

Figure No 3. <https://www.fastcompany.com/90456312/paris-mayor-has-a-dream-for-a-15-minute-city>

12. Finally, the application of a robust monitoring system and a portfolio of actions for climate change are required to know the environmental impacts of this type of urban transformations. Since the VN strategy did not include a consistent monitoring system, the technical assistance helped the city to develop an Environmental Impact Assessment to evaluate the results of the project at the five pilot sites (San Felipe - Juan XXII, Barrios Unidos; 20 de Julio-San Blas, San Cristóbal; El Porvenir, Bosa; Las Cruces-San Bernardo, Santa Fé, and San Carlos, Tunjuelito). Likewise, considering the needs identified by the District Environment Secretariat (SDA) for the San Carlos pilot in the Tunjuelito neighborhood, the WB team developed a portfolio of actions and project sheets for the implementation of mitigation and adaptation actions in that neighborhood.

13. The evaluation of the project results brought impressive impacts and achievements to light. For the San Felipe pilot, the monetary valuation methodology found that every dollar invested in tactical urban interventions has returned USD 15 in health benefits to the community. Other positive effects included a reduction in the concentration of particulate matter and an improvement in air quality, a decrease in noise complaints by 70 percent between 2019 and 2023, a 12.3 percent reduction in average speed on perimeter roads, and about 47 percent fewer traffic accidents between 2019 and 2023. For the San Carlos pilot, decisive inputs were collected for decision-making in the city, revealing the low tree cover of the neighborhood (3.2 percent) and the percentage of effective public space (0.23 m²/inhab) which is the lowest of any Vital Neighborhood.

14. Estimates on the cost of implementing LCVN Strategy in the 33 Local Planning Units (UPL) assume an approximate value of US\$2,01 billion. For the San Felipe and San Carlos pilot cases, the approximate cost is US\$38,7 million and US\$53,3 million respectively. Additionally, the execution of the works corresponding to the 'Alameda Vital' project (Carrera 22 between streets 74 and 77 and related perpendicular roads) that seeks to create a public space focused on pedestrians and the nature of San Felipe LCVN, would have an approximate value of \$3.6 million. Finally, with the purpose of facilitating resource management, it was considered that the execution of the interventions can be carried out in different stages.

15. The technical assistance provided by the World Bank consolidates multiple efforts (seven consultancies) to secure a comprehensive approach to the LCVN Strategy while understanding both its constraints and opportunities for the city. Such task was possible due to the high-level of commitment and involvement of the city of Bogota which brought together several stakeholders beyond the District Mobility Secretariat (SDM) and the District Environment Secretariat (SDA) initially targeted (see Annex 4). As a result of high-level meetings with city officials, work sessions, co-creative workshops, and focus groups, it was possible to reach a common ground highlighting the importance of carrying out future activities that enable securing the implementation of the LCVN Strategy. In this sense, local authorities were key players not just by recognizing current challenges to reduce GHG emissions or to reach the city proximity model, but also committing to take action now by defining a roadmap that facilitates decision making, intersectoral agreements and financial leverage.

16. This report aims to present the final results of the technical assistance that allows measuring the impacts derived from interventions at a local scale, to trigger replicable and scalable urban transformations in different contexts. This support, developed from LCVN Strategy, reveals the results of the application of a management and governance scheme, urban design guidelines, as well as the measurement of environmental impacts as a possible alternative to consolidate a city approach of proximity and resilient to climate change. Also, presents the results of the San Felipe pilot LVCN case.



3. URBAN TRANSFORMATION POLICIES

17. Differences between programs and policies in the city have blocked the effective implementation and management of territorial interventions. These inconsistencies have led to (i) highly dispersed inter-institutional efforts; (ii) lack of clarity regarding the role of agents and their levels of participation; (iii) the absence of authoritative coordination and decision-making bodies; and (iv) difficul-

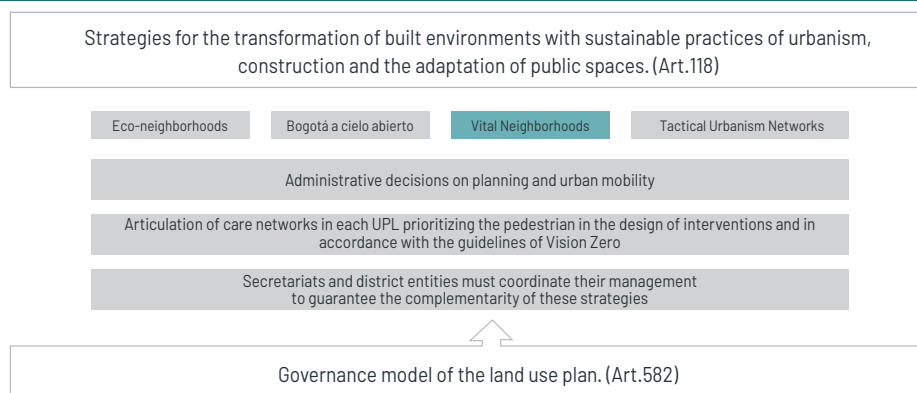
ties ensuring coordinated multi-sector planning, management, and financing. In this sense, it is necessary to propose a 'Roadmap' that serves the city as a tool to achieve intersectoral arrangements, define decision-making instances, and ensure resources that legitimize LCVN Strategy and its sustainability in the medium and long term.

3.1 Legal framework and management model

18. The city's new Land Use Strategy identified key considerations for the preparation and construction of its management model. These considerations included (i) the need for decision making mechanisms related to mobility and urban planning; (ii) articulation of care networks in each

UPL; and (iii) the need for a governance scheme at the district level to oversee inter-institutional coordination in implementing Vital Neighborhoods activities (Figure 4). In relation to the latter, the 'POT Governance Model' (Art. 582) constitutes a viable vehicle for the adoption of the new model.

Figure 4. Legal framework for Vital Neighborhoods Strategy (Decree 555 of 2021)



Sources: JFP & Asociados 2022.

19. One of the first key findings when developing a viable management model for the Land Use Strategy consists of identifying specific phases, which could be both flexible and adjusted over time. As a result, five phases were prioritized as it follows:

- **Planning.** Preparation of the diagnosis for each neighborhood, identifying eco-urbanism⁹ components, actors, triggering interventions, and possible sources of financing. The actors may be either permanent or temporary VN strategy managers, to be determined during the development of

the process, depending on the characteristics of the neighborhood, the implementation plan, and the components to be executed.

- **Structuring.** Prioritization of actions, structuring of the intervention for recovering public spaces,¹⁰ definition of the monitoring requirements for territorial actors, and identification of results and impact indicators.
- **Implementation.** Execution of the pilot and impact evaluation to determine whether the vital neighborhood can be considered a Comprehensive Project¹¹. If the evaluation so

⁹Set of decisions and actions for planning, development, construction, and adaptation of cities, oriented towards sustainable development, to minimize environmental impacts where the relationship between the natural and built environments is balanced and their infrastructures use their resources safely and efficiently.

¹⁰Known as tactical urbanism, this is a collaborative process to recover public space and maximize its shared value. It is carried out through light, low-cost and rapid interventions to explore alternatives to improve the spaces.

¹¹It is an urban intervention instrument that encompasses the dimensions of the physical, social and institutional with the aim of solving specific problems in a defined territory, seeking to improve the living conditions of its inhabitants. Retrieved from: <https://www.sdp.gov.co/transparencia/informacion-interes/glosario/proyectos-urbanos-integrales>

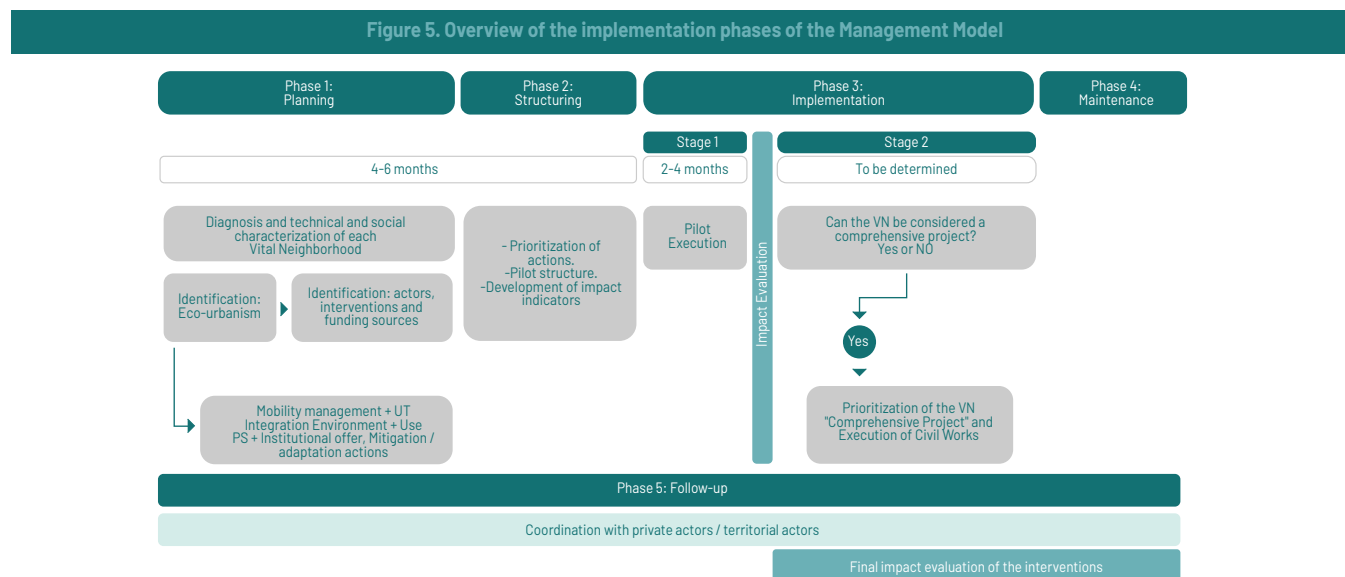
determines, the project can be linked to the necessary administrative support and financing and proceed with the contracting and execution of civil works. If the evaluation determines that the intervention does not have sufficient impact, it will not be eligible for financing for civil works.

- **Maintenance.** Two alternatives for maintenance of the Management Model are proposed: (i) the use of funds generated from activities in public spaces; and (ii) by managing resources for maintenance activities as defined in the Revitalization Plan and the Neighborhood Plan.
- **Evaluation.** Analysis of monitoring indicators to quantify and evaluation the results of obtained from implementation of the LCVN Strategy. It is proposed that the monitoring of the LCVN Strategy in each neighborhood be the responsibility of management or be carried out as part of the Convergence Scheme.

20. Some of the aspects considered by the management

model are mentioned below: the identification of actors about their capacities, functions and roles established within the governance scheme,¹² institutional coordination conditions, and trigger interventions in the priority neighborhoods such as including improving public spaces, creating safer streets, and activating underutilized areas, among others.

21. The analysis found that for a land use management model to be effective, it needs to have clear governance arrangements and decision points. The Bogotá model includes an Institutional Convergence Scheme that sets parameters for inter-institutional coordination, monitoring all phases of implementation (Figure 5), and prioritizing investments. The model also includes management decision points pertaining to the planning, management, and financing of the Land Use Strategy; coordinating the programming of multisectoral actions; and presenting investment needs to the municipal government at the appropriate time in the budget cycle.



Sources: JFP & Asociados 2022.

22. The land use management model for the LCVN pilot in San Felipe determined the actions that would be needed to consolidate the LCVN strategy and scale it up to other areas of the city. That analysis found that for more neighborhoods to evolve toward the low-carbon model, the LCVN strategy needs to be fully incorporated into the Land Use Strategy, which in turn should be incorporated into the District Development Plan. Further, the Convergence Scheme should allow for the participation of the District Mobility Secretariat (SDM) and other relevant departments in

low-carbon mobility schemes.

23. From the management perspective, the LCVN strategy needs to be understood as a comprehensive project that is an integral part of the overall Land Use Strategy. Considering the LCVN in this way will enable the use land management instruments such as Urban Renewal Projects for Sustainable Mobility (PRUM), Modal Integration Areas (AIM) and Strategic Actions (AE) to leverage expertise and financing.

¹²This study, plus the results of the analysis of the San Felipe case, allowed the entities to be classified as 'permanent', 'transitory' or 'anchor' actors. Some agents could act as managers of the strategy and others could be part of the proposed institutional convergence scheme.

24. The use of these financing and value capture instruments can also help to improve and maintain the public spaces provided for the LCVN strategy. The revitalization of public spaces could also be supported by the Special Districts for Improvement and Sectoral Organization

(DEMOS), and by the Bogota Open Sky 2.0 (BACA 2.0¹³), and Bogotá Productive 24 Hours initiatives, which promote productive and cultural activity at unconventional hours through alliances with community organizers, restaurants and merchants.

3.2 Sustainable and secure mobility plan

25. The lack of updating planning instruments such as the Master Mobility Plan hinders the development of policies, strategies, programs, projects, and goals related to the urban growth. Bogotá adopted its first Master Mobility Plan in 2006 (Decree 319), however, given the changes and transformations of the city an update was required. In response, the city has decided to formulate the Safe and Sustainable Mobility Plan (PMSS) in articulation with the proximity city vision established in the POT. It's under this context the Bank accompanied the SDM by reviewing international strategies and successful case studies whose objectives go beyond the concept of mobility and address territorial occupation models, climate change, public space, housing, and health, among others.

26. The Sustainable and Safe Mobility Plan (PMSS) is supporting a sustainable, safe, equitable and inclusive mobility system. Considering territoriality, climate change, public space, housing, health and well-being, the analysis found that the PMSS approach needs to give more attention to addressing (i) the cross-cutting issues of climate change and gender; and (ii) the key land use concepts of Vital Neighborhoods and Care Blocks (MdC). Further, the PMSS monitoring system requires performance indicators that are specific, measurable, achievable, realistic and up to date. These indicators must consider (i) multiple levels (regional, city and local) in the territorialization of the system; and (ii) rural to urban mobility for passenger and cargo transport.

27. The main key policy recommendations suggested to the PMSS are: (i) from an institutional perspective, it is necessary to articulate both the VN Strategy and Care Blocks established within the POT with the PMSS to secure the proximity city model can be viable in the future; (ii) regarding the PMSS approach, focus was very much placed on equity, however, it was suggested to strengthen the climate change and gender dimensions; (iii) for the objectives, it was recommended to direct them towards accessibility multidimensional concept rather than mobility which was restricted by sectoral approach; (iv) regarding the performance indicators, the analyzes focused on the importance of these to be 'SMART': specific, measurable, attainable, realistic and timely. At the same time, it was suggested that they could be better linked to each of the strategies described by the plan; (v) spatially, it was necessary to strengthen the multi-scale territorial approach e.g., regional, city, local; as well as defining a unit or polygon of action for both indicators and strategies; and (vi) a significant reduction (30 percent approx.) in programs (80) and projects (120) was reached in order to avoid technical redundancies and inefficiencies when implementing them in future stages.

¹³According to Decree 070 of 2022, which adopt the public space revitalization strategy known as "Bogotá a Cielo Abierto 2.0", and regulate the economic use of public space in the Capital District associated with said strategy. Retrieved from: <https://www.alcaldiabogota.gov.co/sisjur/listados/tematica2.jsp?subtema=36092&cadena=#:~:text=Decreto%20070%20de%202022%20Alcald%C3%ADa,Capital%20asociadas%20a%20dicha%20estrategia.>



4. URBAN DESIGNS FOR TRANSFORMATION

28. A shift towards a new urban paradigm urges to reverse Bogotá's current unsustainable growth model.

During the last 20 years, the city considerably increased its urban footprint by 15.34 percent (IDOM, 2016) leading to a threat in its urban ecosystems and the viability of the city. The proximity city paradigm reverses such growth patterns by prioritizing local scale while encouraging 'neighborhoods' to emerge as essential urban piece. Considering the new

distribution of 33 UPLs to ensure greater territorial balance, understanding the principles of Ecosystemic Urbanism becomes essential to continue consolidating a comprehensive toolkit that serves territorial decision making. As a result of the technical assistance provided by the World Bank to the city, the consultancies focused on preparing both 'Urban design guidelines' and 'Implementation guidelines' for the Vital Neighborhoods Strategy.

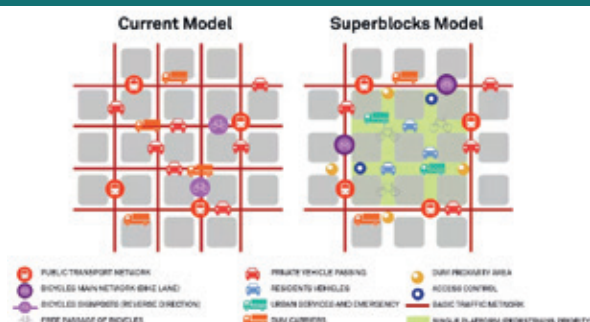
4.1 Guidelines for urban transformation in the framework of low-carbon neighborhoods

29. To territorialize ecosystemic urbanism, the study proposes a superblock model for the organization and management of public spaces.

In Barcelona, the superblock model has made neighborhoods more comfortable, walkable and liveable, with more community interaction and less traffic, and with proximity to public spaces, nature, transport and basic necessities (Figure 6). This model incorporates: (i) a new territorial unit conceived as a new organization cell for the different pieces of which the city is composed and (ii) a piece that allows to establish the functional model (organization of mobility networks in their operation and management) and urban planning (physical and definitive transformation of public space and the

environment) in an integrated way under an essential ecosystem vision for territorial planning.

Figure 6. A Superblock in Barcelona



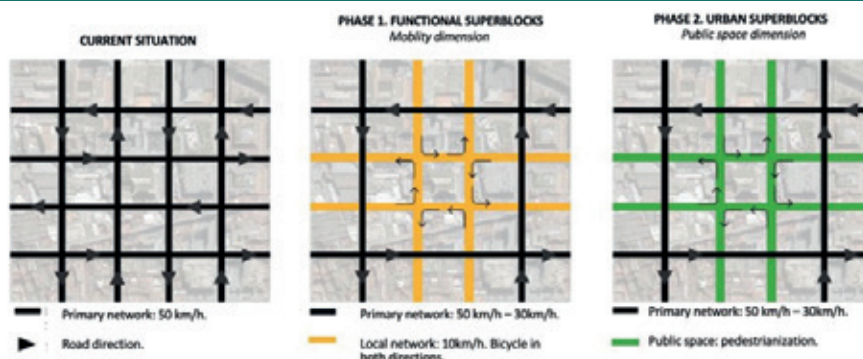
Source: Salvador Rueda, 2022

30. In Bogotá, the proposed superblock model includes an analytic approach for its development that aims to:

(i) Promote the compact, proximity and mixed-use city model that favors more sustainable mobility patterns; (ii) establish modal distributions favorable to the most sustainable modes of transport; (iii) reduce the impact of motorized travel and increase opportunities for alternative, lower-impact modes

of transport; (iv) provide for local public spaces that encourage leisure, participation, and relationships with neighbors; (v) prioritize the creation of walkable areas to increase well-being and revitalize commercial activity; (vi) increase the biodiversity and connectivity of urban green spaces; and (vii) promote local economic activity and social innovation.

Figure 7. Mobility rearrangement phases of the Superblocks model



Source: Urban and Territorial Ecology Foundation, 2023

31. The identification of dimensions of analysis were considered for the development of proposals consistent with the specific conditions of each territory: (i) urban and socioeconomic, associated with urban morphology and structure and sociodemographic profile; (ii) mobility, which addresses how the population moves (walking, cycling, public or private transportation), infrastructure and services

(road network, parking, urban distribution of facilities), accessibility and road safety; and (iii) public space and green infrastructure which considers its structure (use, quantity and connectivity) as well as habitability in terms of ergonomics, comfort and attraction. Table 1 details the general methodology for each dimension and the tools applied for the management and processing of information.

Table 1. Ecosystemic Urbanism and Superblocks model

 Dimensions	 Components	 Variables	 REORDERING CRITERIA (superblocks proposal)
Urban and Socioeconomic Area	Morphology and urban structure	- Population and housing density - Urban compactness - Activities on the 1st floor - Social and local facilities	
	Sociodemographic profile	- Age and gender - Population mix - Population income	
Scope of Mobility	Modes of mobility	- Mobility on foot, bicycle - Public transport - Private vehicle - Vehicle management plan	- Prioritization of the road network - Reorganization of the networks for the different transport modes and services
	Road network and mobility services	- Parking lot - Urban distribution of goods - Accessibility and road safety	- Parking regulation - Urban distribution of goods - Road safety
	Gender-based mobility		Gender-based mobility
	Externalities of the current model	- Accident rate - Air quality and noise - Functional appropriation of public space	
Scope of Public space and green infrastructure	Structure of public space	- Uses of public space - Living space per inhabitant - Green areas - Green connectivity	Identification of attracting and generating areas of activity
	Public space habitability	- Degree of ergonomics - Degree of comfort - Degree of attraction	- Degree of ergonomics - Degree of comfort - Degree of attraction

Source. Author, based on Salvador Rueda, 2022.

32. The Superblocks model seek to turn neighborhoods into walkable and meeting places where expression and exchange between inhabitants occur. Likewise, they must enjoy comfortable conditions (in terms of environmental and auditory contamination and temperature), must be accessible to all and enjoy a good relationship with the buildability of each context. Through the pacification within the Superblocks¹⁴, the accessibility of sustainable modes and the flow of bicycles and pedestrians is prioritized.

¹⁴ Through the implementation of speed reduction elements on the interior roads, such as chicanes, narrowing, the implementation of street furniture and horizontal signage on the roads.

4.2 Case studies for the implementation of urban design guidelines

33. The Implementation Guidelines seek to align the expansion of Vital Neighborhoods Strategy with district policies and the superblock model. The guidelines lay out three axes of strategic action: (i) reorganization of mobility in favor of sustainable modes; (ii) revitalization of public space, and green and resilient infrastructure; and (iii) strengthening of infrastructure for public facilities, considering the basic needs of the people through the reduction of motorization, the level of air pollution and acoustic impact, and the level of occupation of the road space.

34. The guidelines were developed based on analysis of the LCVN pilots in San Felipe, San Carlos, and Santa Fé. The analysis considered: (i) the characteristics of each neighborhood; (ii) lessons learned from the pilot phases; (iii) each district's proposed master plan for expanding the VN model; (iv) preliminary costs; (v) implementation vehicle (management model) and possible institutional arrange-

ments; and (vi) citizen mobilization and communication strategy.

35. Preparation of the guidelines began with a review of the key planning instruments for the LCVN strategy developed by the city. The review included: (i) codify the regulatory foundations of the LCVN Strategy within the Land Use Plan (Decree 555 of 2021); (ii) the Technical Support Document (DTS) for the LCVN strategy issued by the District Planning Secretariat (SDM); and (iii) the specific Technical Support Document for each prioritized neighborhood. The review also considered the degree to which the pilots achieved four main objectives: (i) prioritization of sustainable transport modes; (ii) resignification of the use of neighborhood streets; (iii) speed management; and (iv) social engagement and citizen participation. Case studies on the San Felipe and San Carlos pilots and expansion proposals are presented below.

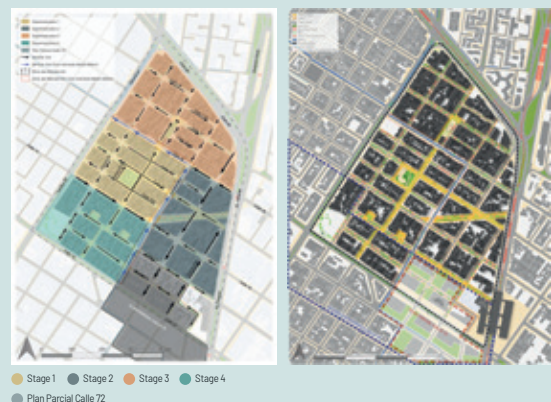
Box 1. San Felipe Vital Neighborhood

The San Felipe neighborhood is part of the Los Alcazares Zonal Planning Unit (UPZ) in the Barrios Unidos district. The 37 ha area is bounded by 80th Street to the north, 72nd Street to the south, Avenida Caracas to the east and Carrera 24 to the west.

In 2021 and 2022, the SDM developed the first two stages of the LCVN project (the pilot phase) in San Felipe (see map, left side). The pilot consisted of diagnosis, planning and implementation of Tactical Urbanism actions (i.e., exploration of low-cost and rapid interventions to expand and improve the quality of public space) and allow for the implementation of superblock principles.

The pilot phase ended in 2022 with the installation of modal filters in Araña Park and along Carrera 22, at the intersections of 75th A, 75th and 74th Streets, forcing motorized vehicles to turn instead using the interior of the neighborhood as a shortcut. These interventions reinforced the superblock principles by creating safe sidewalks and local streets while recovering the intersections for pedestrians and bicyclists.

Figure 8. Conceptual urban design proposal – Superblock model in San Felipe VN



Proposed approach to scaling up

Following the pilot phase, the SDM proposed a master plan to scale up the recovery of public space throughout San Felipe (see map, right side). The proposal includes infrastructure and urban renewal projects that have the potential to (i) create new hubs for economic activity and sustainable transport; and (ii) maximize community interaction and cohesiveness by reallocating about 60 percent of street space to pedestrian circulation and meeting points.

Stages for the execution of San Felipe's Vital Neighborhood

- **Stage I – West Superblock**, from Cra. 20c to Cra. 24 and from 74th to 76th Street began with the installation of modal filters around La Araña Park, forcing traffic to circulate around the park and allowing for a high concentration of cultural activity. These interventions created the city's first superblock.
- **Stage II – East Superblock** prioritized 74th Bis Street as a pedestrian and environmental connection axis between Caracas Avenue (TransMilenio and Metro Line 1) and Superblock 1, further consolidating the West Superblock as the cultural heart of the neighborhood.
- **Stage III – North Superblock** seeks to reduce the number of motorized vehicles on interior roads, create more parking areas and continue to prioritize pedestrians along the axis of Cra. 22.
- **Stage IV – South Superblock** seeks to complete the mobility scheme while connecting two green areas on Cl. 73 with the pedestrian priority axes of Cra. 73 and Cra. 22.

Alameda Vital San Felipe (Pre-Feasibility studies Carrera 22 between the 74 and 77 Streets, and adjoining perpendicular streets)

One of the prioritized projects for San Felipe are the permanent works of Carrera 22 between 74 and 77 streets. The objective is to create a public space focused on pedestrians and nature for the Vital San Felipe neighborhood in which its components are effectively integrated and articulated to eliminate the hierarchy of the vehicle (less than 40% of the area) over other actors of the public space such as pedestrians or nature.

The specific objectives are (i) Consolidation of the public space identity; (ii) Increase in the space set aside for nature; (iii) Strengthening pedestrian protection; (iv) encourage the appropriation of public space; (v) Provide universal accessibility.

The selected alternative (among 3) was "Alameda Vital". The concept was guided by the environmental and biotic component of the public space, it proposes the re-naturalization of the neighborhood through the integration of permeable landscape areas and sustainable urban drainage systems (SUDS). The Juan XXIII and la Araña parks are connected through the boulevard on Carrera 22, generating a larger-scale urban piece with the potential to be more easily linked to the environmental structure of the city.

The proposal, in terms of pedestrian circulation, has an increase of 2,948m² equivalent to 20 percent. Regarding nature areas, which include landscape and sustainable urban drainage systems areas, there is an increase of 2,134m² equivalent to a 15 percent increase, exceeding 10 percent of the project area. Vehicular circulation decreases in area 5,303m² equivalent to 36 percent less area allocated to the vehicle.

Figure 9. Rendering of Alameda Vital San Felipe



Source: Taller Architects. San Felipe pre-feasibility design (2023)

Box 2. San Carlos Vital Neighborhood

The San Carlos neighborhood belongs to the Tunjuelito 62 Zonal Planning Unit (UPZ) in the Tunjuelito locality. The 61 ha area is bounded by Diagonal 48 South to the north, 58th South Street to the south, Caracas Avenue to the east and Carrera 19C to the west.

In late 2022, as part of the conceptual design proposal under the San Carlos Vital Neighborhood Strategy by the District Mobility Secretariat (SDM), it was initiated the demarcation of some critical points within the neighborhood, such as the intersection of Diagonal 54 South and Cra. 18B, with the intention of piloting some road safety concepts.

The pilot was based on the superblock principle of preventing motorized vehicles from using the interior of the neighborhood as a shortcut. It was also intended to reduce the cost of road maintenance and improve circulation around important local landmarks such as the San Carlos Market, while discouraging through traffic. The use of modal filters along Diagonal 54 South aimed to increase pedestrian space and consolidate the vision of a new Green Axis for San Carlos.

Proposed approach to scaling up

This design concept focused on: (i) providing access to commercial roads; (ii) improving the quality of green areas along the public road space; and (iii) creating meeting points with green corridors and public squares.

After changing the traffic flow at the intersection of Diag. 54 South and Cra. 18, SDM then tested the narrowing of lanes and widening of sidewalks to create safe crossings for pedestrians. The pilot maintained the Integrated Public Transport System (SITP) routes as a priority for circulation along Diag. 54 South. In addition, the pilot focused on (i) tactical actions close to the San Carlos Market and the parks to the north; and (ii) low-cost interventions around the neighborhood where pedestrians could connect to the periphery, as a first step toward implementing a broader superblock scheme.

Figure 10. Conceptual urban design proposal - Superblock model in San Carlos VN



Source: Implementation and Design Guide for LCVN

The creation of superblocks in San Carlos

• **Stage I – San Carlos Green Axis Superblock:** executed the Green Axis on Diagonal 54 South. This intervention is delimited from Carrera. 19a to Caracas Avenue to: (i) allow circulation on 55 South Street without obstruction; (ii) generate a connection on Caracas Avenue at one of the points with the greatest number of road accidents, by reducing vehicle speed; and (iii) create two main squares for the neighborhood at the intersection of Carrera. 18b and Caracas Avenue

• **Stage II – South Superblock:** prioritizes the continuation of the Green Axis along 55 South Street between Cra. 19a and Carrera. 19c to make the connection with the Metropolitan Park El Tunal. Additionally, one of the main roads, 55 South Street, will be intervened with a pacification treatment that places it in a one-way road and with additional spaces for on-street parking, loading/unloading of goods, improved sidewalks for circulation and landscaping. Additionally, priority is given in Stage II to the south side of the San Carlos neighborhood, given the Strategic Actions (SA) taking place in the San Benito and Tunjuelito neighborhoods, as part of the urban renewal strategy of the Tunjuelito UPL established by SDP.

• **Stage III – North Superblock:** seeks to reinforce the San Carlos Market Square, which has scheduled interventions to improve the immediate public space by the Road Maintenance Unit. It also seeks to strengthen the existence of El Planchón Park on the northern edge, which is one of the main local parks. It is recommended to intervene along 53 South Street, taking advantage of it as a commercial axis.

Figure 11. Green axis proposal at Diagonal 54 South



Source: Implementation and Design Guide for LCVN (2023)

36. According to the analysis developed, the following results were found: (i) the estimated cost of the total implementation of the 33 LCVN is US\$ 2,01 billion¹⁵. For the implementation of Phase I (Tactical Urbanism) the estimated cost is approx. US\$370 million¹⁶; (ii) the estimated cost to finish the implementation of San Felipe is US\$38,7 million, the cost for the construction of 'Alameda Vital' is US\$3,2 million¹⁷; and (iii) the estimated cost for the implementation of San Carlos is US\$53,3 million (Annex 1).

37. Within the scope of the conceptual design developed for LCVN, preliminary indicators¹⁸ were determined in the areas of Mobility and Public Space to measure the impact of the three Vital Neighborhoods. The projects include changes to the public space and are related to metropolitan-scale urban renewal actions that will have an impact on land value and possible gentrification. However, the development of an impact study is suggested in later

stages of pre-feasibility and feasibility in relation to these aspects of the implementation of the LCVN.

38. In the case of San Felipe, a real estate development will occur with the construction of the two new metro lines. This situation generates a real state and urban development incentive to densify other areas of this neighborhood. In terms of urban development processes, once the project has been developed and is in operation, there may be an interest of the external neighbors to prolong the spaces and propose new development ideas based on the new experiences. As a real estate process, it could be presented in the interest of adapting, reforming, or renovating private spaces to increase the services offered and the capacity of people.

39. In addition to the above, a new framework where activities such as -Bogotá Cielo Abierto 2.0- (economic use) or -Bogotá 24 hours- can be implemented is being

¹⁵To estimate the total cost of the 33 Vital Neighborhoods program, several assumptions are made:

(i) The total intervention area of the five projects that the Mobility Secretariat has already analyzed - San Felipe, San Carlos, Las Cruces, Bosa, and La Victoria - is averaged. It is assumed that the area of the other 28 Vital Neighborhoods will be similar to the areas studied, so an average of these areas is used; (ii) An indicator of value per hectare is estimated based on the costs for the San Felipe and San Carlos projects in this consultancy, and it is assumed that the cost per hectare in all Vital Neighborhoods will be an average of these two; (iii) Costs for the three analyzed Vital Neighborhoods were estimated according to the proposed conceptual designs. However, it is worth noting that San Felipe and San Carlos were approached as infrastructure projects while Historic Center is developed as a tactical urbanism pilot. Thus, these preliminary costs shouldn't be compared between them as they are indicative of two different types of projects; (iv) The preliminary costs for the execution of the works of the conceptual design were elaborated based on the estimated 2022 reference values of the IDU corresponding to the right-of-ways for the city of Bogotá, assuming adjustments for the change of validity to 2023; (iv) The adjustments correspond to reports from DANE on the IPC for the first quarter of 2023 with an indicator of 13.12%. Likewise, the adjustment of 16% on the Minimum Legal Wage in force was taken into account for the development of the exercise.

¹⁶The scenarios assume finishes for public spaces with more durable materials such as hydraulic concrete, as well as the undergrounding and updating of energy, sewage, water supply, and other networks.

¹⁷General lengths and areas were quantified for the main chapters of the project. The unit prices were taken from the IDU price list. A % of AIU of 50% is considered, which includes for this evaluation, the general costs of work, personnel. IUS\$ = COPS4.430 (18 April/2023)

¹⁸It is worth noting that not all indicators are measured for the proposed LCVN in the Historic Center because the conceptual proposal has a scope of tactical urbanism where infrastructure is not intervened. Therefore, it is necessary to design and estimate the impacts on new green areas, areas of permanence (plazas), among others, in a later stage.

created within the city. This offers an example of a methodology that can be implemented in other areas, not only as a Vital Neighborhood of the SDM but articulated with other initiatives of the Mayor's Office. This also includes the Real Estate Developers of the Metro, which can be part of the urban charges to provide financing for the implementation of the Vital Neighborhood in other San Felipe areas or finance part of the construction of prioritized projects in the medium term with real estate developments towards the Metro lines.

40. management model proposed consists of five main dimensions: (i) Planning and citizen participation; (ii) inter-institutional coordination; (iii) applicable regulations; (iv) resource management; and (v) coordination with territorial actors. This management model responds to the request of the SDM to the WB and is responsible for the successful implementation of the Vital Neighborhoods program. Based on an analysis of competencies and actors, that management be assumed by the SDM and the SDP jointly. However, the Model would benefit itself from specific articulation with other secretaries such as District Habitat Secretariat (SDH) and District Woman's Secretariat (SDMujer) who lead the 'District System of Care' through 'Manzanas del Cuidado'. It is worth mentioning that the five dimensions of the management model are contained within the Management Office which was designed for interinstitutional and

strategic projects such as Vital Neighborhoods, and whose main function is the articulation between actors at the inter-institutional level, the union between short-term visions, and monitor the consistency of these actions with the POT.

41. In accordance with the above, the implementation of the management model for each vital neighborhood may change according to the ongoing projects in the area and the nature of the design. Thus, for the Vital Neighborhood San Felipe, the Culture Secretary (SDCRD) and the Economic Development Secretary (SDE) are key players that should be involved, given its Creative District status and the focus on promoting local businesses. For Vital Neighborhood San Carlos, the Road Maintenance Unit and Aqueduct Company are recommended given the focus on public space improvements to the San Carlos Food Market let by the former agency, as well as the projects happening around the catchment area of the Tunjuelo River, next to the project area, led by the former. Finally, for Vital Neighborhood Historic Center, the District Woman's Secretariat (SDMujer) and the Heritage Institute are key players given the ongoing Manzana del Cuidado project, as well as the Historic Center Revitalization Project.



5. MEASURING THE ENVIRONMENTAL IMPACT OF URBAN TRANSFORMATION

42. By 2023 the LCVN Strategy has prioritized tactical urbanism intervention in the five neighborhoods identified as pilots. The selection of these pilots responded to a territorial characterization exercise led by SDM, regarding the constraints and opportunities as well as the potential for replication or scaling up to other areas of the city. Considering the scope of the technical assistance provided by the WB, two neighborhoods were selected applying differential approaches according to their specific conditions.

43. This chapter is divided in two sections. The first one applies an Environmental Impact Assessment Methodology to explore the benefits derived from interventions in the urban environment in San Felipe VN. San Felipe's strategic location benefits from its proximity to major mobility corridors of the city, and it has a cultural vocation which

brings together multiple stakeholders. It also represents the first pilot defined by the Mayor's Office, being the area with the greatest progress facilitating a baseline of robust indicators to carry out a specific results measurement.

44. The second section recognizes both environmental threats and potentials of San Carlos VN and defines a Portfolio of adaptation and mitigation actions to tackle climate change. The prioritization of this neighborhood attended to two considerations: (i) imminent threats regarding feasible loss of ecosystems in its influential area; and (ii) green infrastructure deficits within the neighborhood and heat island effect. Finally, the spatial urban configuration of the neighborhood was also pondered and favored as it represented ideal conditions for decision making regarding the principles of both ecosystemic urbanism and superblocks.

5.1 Environmental Impact Assessment in Urban Design

45. Measuring the environmental impact of urban designs framed by the LCVN strategy is important for several reasons: (i) accountability, to demonstrate transparency while helping to build public trust; (ii) compliance, to ensure that local authorities adhere to air quality, low carbon, road safety and public space policies; (iii) resource allocation toward priority interventions to address the most disruptive environmental changes; (iv) planning, based on an Environmental Impact Assessment (EIA), which can help the district make more informed decisions about the design and location of infrastructure for the next stages of the LCVN.

46. Given the specific characteristics of the tactical urbanism¹⁹ pilot interventions carried out by SDM, a full EIA may not be needed, particularly since other methodologies²⁰ are better adapted to this type of projects. In the case of LCVN strategy, it was proposed to conduct a Rapid Environmental Impact Assessment (REA) with focus on health impacts derived from lowering air pollution, noise and GHG emissions, as well as from increasing physical activity due to a higher modal share of cycling and walking. Table 2 shows the list of effects considered in this assessment including the unit values and its sources.

¹⁹Temporary, or low-cost interventions to transform urban spaces quickly and with minimal resources

²⁰Rapid Environmental Impact Assessment (REA), that involves a rapid assessment of the environmental impacts based on a set of standardized criteria; Sustainability Impact Assessment (SIA), for focusing on the broader sustainability impacts of a project, including economic and social impacts in addition to environmental impacts; or Participatory Impact Assessment (PIA), that implies the participation of stakeholders including residents and community groups

Table 2. List of effects considered in LCVN EIA with focus on public health

Effect	Unit Value	Units	Sources	Main Assumptions
Green House Gases mitigation	\$ 434	USD/ton CO _{2e}	Ricke et al., 2018b	Average value of the social cost of carbon, globally
Local Air Pollutants mitigation	\$ 145.091	USD/ton PM _{2.5}	Castillo et al., 2010 y Sefair et al., 2019 Korzhenyevych et al., 2014 US EPA, 2013	Analyzes that include morbidity and mortality benefits in terms of direct health costs, lost productivity, and statistical value of life
Noise Reductions	\$ 0,003	USD/VKM	Gössling et al., 2019b	The value considers direct health costs, lost productivity, the cost of discomfort and reduced property values
Health benefits from walking	\$ 0,402	USD/PKM	Gössling et al., 2019b	Private and social health benefits from increased physical activity, direct savings to the health system, savings from loss of productivity, savings from life extension. These associated with reduced risk of cardiovascular disease, various types of cancer, diabetes, depression, and obesity
Health benefits from biking	\$ 0,223	USD/PKM	Gössling et al., 2019b	

Source: Hill Consulting, 2023

47. The monetary valuation of the expected benefits is based on secondary information obtained from local and international studies. However, the methodology used here can provide a useful starting point for assessing the impacts of these interventions and advocating for their replication. The evaluation of each effect was based on unit values shown in Table 2, which associate each effect with a monetary benefit (e.g., dollars per kilometer, dollars per passenger, dollars per ton of pollutant). Information from other countries was used in cases where no information was found for Colombia, in which case the benefit transfer methodology was used. It is important to mention that road safety improvements, green spaces, urban gardens, and crime reduction could also be monetized with additional research.

48. Based on the monetary valuation methodology, the EIA conducted for San Felipe found significant benefits from the proposed changes in mobility patterns. The calculations used mobility data from the Origin-Destination household surveys (2019), stated preferences interception surveys carried out by SDM (2020, 2022), and PM2,5 and CO2eq emission factors for each transport mode. These data showed that environmental benefits from changes in mobility patterns amounted to USD 3.16 million between 2020 and 2022. According to the findings, every dollar invested in tactical urban interventions has returned USD15 in health benefits to society. Table 3 below shows the calculation of benefits for air pollution and GHG mitigation during the same two-year period. It also shows the benefits from noise reduction and increased physical activity from changes in mobility patterns.

Table 3. Economic benefits due to changes in emissions and activity for each transport mode

Transport Modes	Air Pollution and GHG			Noise Reduction and Physical activity		
	Change in PM _{2.5} (Ton)	Change in CO _{2eq} (Ton)	USD	Change in Activity (pkm*/vkm)	USD	
Cycle	0,00	0	\$ -	3.573.928	-\$	797.341
Pedestrian	0,00	0	\$ -	380.361	-\$	153.010
Motorcycles	0,00	-2	-\$ 688	-23.293	-\$	70
Taxi	-0,01	-671	-\$ 292.505	-3.201.305	-\$	1.287.808
Bus - Transit	0,06	787	\$ 349.669	751.079	\$	167.565
Automobile	-0,03	-2.621	-\$ 1.142.169	-10.973.754	\$	-
Totals	0,01	-2.506	-\$ 1.085.692	-9.492.984	-\$	2.070.664

Source: Hill, 2023

²¹ The transfer of unit values between two countries is based on the difference between the two countries in terms of GDP adjusted for purchasing power parity

Monitoring system and baselines: Paving the way for an exhaustive and comprehensive EIA

49. In addition to the REA, a baseline estimation was established prior to developing a more comprehensive EIA for the LCVN strategy. Baselines provide a reference point against which the impact of the intervention can be assessed. From January to April 2023, SDM received technical support for the development of a Monitoring System for the LCVN strategy (Annex 2). The support included (i) defining, designing and prioritizing environmental indicators (air quality, GHG emissions, visual and noise pollution); (ii) providing the calculation methodology and measurement times for each indicator; (iii) proposing complementary indicators and their respective methodologies; (iv) compiling existing information, calculating the baseline indicators and defining follow-up indicators for five Vital Neighborhoods; and (v)

providing a summary of the main impacts and results of the study that can be used to develop communication materials targeted to the general public.

50. SDM received several outputs from the technical assistance process, including: (i) a database with baseline indicators for five Vital Neighborhoods and follow-up indicators for San Felipe's pilot; (ii) geographic database and geographic information processing algorithms for calculating indicators; and (iii) an indicator visualization dashboard in Power BI® (Figure 12 below) linked to the database and able to be updated in the coming years. **The Dashboard is available at this link.**

Figure 12. Monitoring system dashboard - PowerBI®



Source: Hill Consulting, 2023

²²Without a baseline, it would be difficult to determine the extent to which the intervention has affected the environment, making it difficult to identify any potential negative impacts or to measure the success of the intervention

²³San Felipe-Juan XXII (Barrios Unidos), 20 de Julio-San Blas (San Cristóbal), El Porvenir (Bosa), Las Cruces-San Bernardo (Santa Fé) and San Carlos (Tunjuelito).

51. The monitoring system and baseline calculations will allow local authorities and project planners to: (i) measure the impact of the intervention and identify any unintended negative consequences; and (ii) inform decision making about the appropriate design for each location; and (iii) ensure that future planned interventions minimize negative environmental impacts and maximize positive outcomes.

52. The baselines and available information were different for each Vital Neighborhood, especially regarding measurements of air quality and noise. For San Felipe, the indicators suggest that the interventions had a number of positive effects, including a reduction in air pollution, a decrease in vehicle operating speeds, a change in the modal share of transit toward more transport (walking, bicycles), a lower incidence of conflicts over the use of public space, and greater commercial dynamism in the area. These changes correspond with the district's objectives of (i) prioritizing sustainable modes of transport; (ii) opening the streets to new uses; (iii) managing speed; and (iv) promoting economic development.

53. Some of the most relevant findings for San Felipe and other Vital Neighborhoods related with the indicator's quantification exercise are listed below:

- **The concentration of particulate matter (2,6 $\mu\text{g}/\text{m}^3$ for PM_{10} at baseline) was reduced.** Using the methodology proposed by the Agència de Salut Pública de Barcelona, the differences were compared with the measurements of the fixed station of the Air Quality Monitoring Network of Bogotá de Las Ferias. This allowed for verification of a statistically significant change due to the LCVN intervention with a p value of 0.0205 given a two-tailed t-Student test.

- **Between 2021 and 2022, there was a decrease of close to 20 percent in the number of days that exceeded the $\text{PM}_{2.5}$ daily standard (15 $\mu\text{g}/\text{m}^3$).** In contrast, considering the information available during the first quarter of 2023 from the El Provenir and Las Cruces microsenors, $\text{PM}_{2.5}$ concentrations greater than 15 $\mu\text{g}/\text{m}^3$ were identified for more than 40 percent of the days for which there were at least 18 hourly reports. This indicates a potential for air quality to adjust to the reference levels established by the World Health Organization. **(2021).**

- **Noise complaints in San Felipe decreased by close to 70 percent in 2023 compared to 2019,** although more data may be required to validate this trend. While noise

levels in the 20 de Julio-San Blas and El Porvenir neighborhoods began to fall in 2020, complaints in San Carlos increased in 2020, largely due to activities in the industrial sector.

- **On the inner streets of San Felipe, there was a 6.6 percent reduction in average speed** between 6:00 am to 8:00 pm between 2021 to 2022, **from 17.3 km/h to 14.3 km/h.** On perimeter roads, there was a 12.3 percent reduction in average speed, from 28.4 km/h to 25.1 km/h.

- **There were about 47 percent fewer road accidents in 2023 compared to 2019.** However, there was about a 30 percent increase in the number of serious accidents (with injuries or deaths) between 2021 and 2022, going from 27 to 37 registered cases. This lack of correlation between the decrease in speed and the incidence of accidents suggests the need for a detailed case-by-case analysis to identify possible infrastructure improvements.

- **According to the information on tree canopies for 2021, the San Carlos Vital Neighborhood had a tree cover of only 3.2 percent** and effective public of only 0.23 m^2/inhab – the lowest of any Vital Neighborhood if it is compared with the desired urban public space indicator of up to 9 m^2 per inhabitant according to WHO.

54. A robust monitoring system would help to demonstrate the positive effects of LCVN interventions and increase awareness of the need to replicate them in other neighborhoods. If rapid, low-cost tactical urbanism impacts are measured and communicated, a budget for additional civil works would be easy to obtain. For these additional works, the SDM should appoint a LCVN work team that would be responsible for (i) regularly updating the indicators based on the defined methodologies; (ii) setting up surveys and traffic counts to feed into the indicators; (iii) allocate resources based on information collected through contracts and agreements managed by SDM, such as the ones signed with *TPD Ingeniería S.A. and Waze®*.

55. It is strongly recommended that SMD make the monitoring system dashboard available on its website, in order to mobilize support for the LCVN interventions. An LCVN database, updated yearly, should also be published on the Bogotá Open Data platform, to facilitate deeper analysis by researchers.

56. Monitoring changes in land and building prices might also be useful to help SDM (i) identify possible capital gains from the interventions; (ii) implement land value capture instruments to promote investment in new civil works; and (iii) prevent gentrification by involving local residents in planning processes for new public infrastructure and private building projects.

57. Considering the likely change in district government in 2024, it is suggested that a formal intersectoral commitment (e.g, a Memorandum of Understanding or Convergence Scheme) be established among SDM, SDA, SDH, SDP, SDDE and SDMujer. The commitment should include sufficient budget and other resources to maintain the momentum of the LCVN program while a new District Development Plan is established under the new mayor.

5.2 Environmental design guidelines and public space

58. The methodology for preparing the climate action portfolio in San Carlos included the active participation of district entities/agencies and civil society, and coordination with other consultancies of this report led by the World Bank. The guidelines were developed over four phases:

- **Phase 1. Diagnosis.** Integration of background information; description (1:5000 scale cartography) of land use, public space, mobility, social and environmental characteristics; and disaster risk analysis of the San Carlos neighborhood and its area of influence. The result of this phase was the identification of 31 critical points in the neighborhood related to climate change, which were grouped into five main challenges: (i) lack of public space; (ii) limited connectivity with the Main Ecological Structure (EEP); (iii) lack of non-polluting transit options; (iv) poor environmental health and basic sanitation; and (v) risk of climatic disasters.

- **Phase 2. Analysis of alternatives.** The five main challenges were reformulated into positive objectives and a proposed Portfolio of 15 Actions for Adaptation and/or Mitigation of Climate Change.

- **Phase 3. Evaluation.** The proposed measures were evaluated using three main criteria from the AHP multi-criteria evaluation methodology²⁴: (i) social; (ii) environmental; and (iii) pre-feasibility. The result was the definition (supported by metrics) of priority six actions to be carried out in San Carlos and its area of influence (Figure 10).

- **Phase 4. Portfolio of prioritized actions and roadmap.** For each of the priority actions, a roadmap was constructed showing its feasibility, defining preliminary costs, and identifying institutional arrangements, measurement indicators and possible funding sources.

59. For San Carlos, a Base Portfolio of 15 Actions for Mitigation and Adaptation of Climate Change was established. The critical points were grouped into five major problems: (i) deficit of spaces for the enjoyment of its inhabitants; (ii) limited connectivity with the Main Ecological Structure (EEP); (iii) deficient and polluting mobility; (iv) Environmental health and basic sanitation problems; and (v) risk of climatic disasters.

60. Based on the roadmap, the six priority actions were developed to conceptual design level (Figure 13). The estimated cost for implementing these six adaptation and mitigation actions in San Carlos and its area of influence is US\$4,1 million (Annex 3).

²⁴In the theory of decision making, the analytic hierarchy process (AHP), also analytical hierarchy process, is a structured technique for organizing and analyzing complex decisions, based on mathematics and psychology. It was developed by Thomas L. Saaty in the 1970s. The AHP combines math and psychology to compare several options and select the best one. It does this by using a concept called pairwise comparisons. Instead of comparing several criteria at once, they are compared two at a time. That way, the choice is easier to make.

Figure 13. Climate Adaptation and Mitigation actions for San Carlos

Action 1. Renaturalization of effective public space and sidewalks: This action aims to (i) increase the green areas of public space; (ii) improve the existing impermeable surface in the neighborhood; (iii) increase urban tree coverage; and (iv) expand the green areas in landscaping strips next to sidewalks.



Action 2. Pedestrianization of roads to increase effective public space: This action aims to address the deficit of effective and pedestrian public space by converting hard areas into green zones.



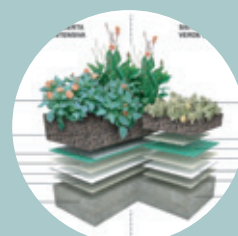
Action 3. Sustainable mobility: Through the implementation of a new sustainable mobility model, the aim of this action is to reduce GHG emissions, discourage motorized vehicles within the neighborhood and recover car space for pedestrians and nature.



Action 4. Increase tree cover: The aim of this action is to increase and maintain tree cover in the neighborhood and its area of influence, in order to (i) aid in carbon storage and sequestration; (ii) improve the connectivity of the neighborhood with nearby elements of the main ecological structure; and (iii) provide spaces to moderate temperature in the neighborhood and reduce the heat island effect.



Action 5. Green infrastructure: This action aims to reduce deficit of green areas in the neighborhood by increasing the number of green roofs and vertical gardens on both private and public buildings.



Action 6. Legality of water connections: The purpose of this action is to eliminate inefficient connections that generate pollutant load in the Chiguaza Creek and the Tunjuelo River, in order to improve water quality and reduce the generation of GHG due to the degradation of organic matter.



61. The analysis led to the identification of actors and agents that would enable the implementation. Three in particular stand out: (i) District Mobility Secretary, which has worked in San Carlos with the Vital Neighborhood strategy, structuring mobility, public space and urban revitalization actions; (ii) the District Environment Secretary which has identified the main needs to be resolved in the neighborhood; and (iii) the José Celestino Mutis Botanical Garden, which operates the Urban Tree Management Information System (SIGAU) containing official information on trees located in the city's public urban space. Secondary stakeholders were identified as: (i) the Mayor's Office of Bogotá; (b) the local government of Tunjuelo; (iii) the District Planning Department, (iv) IDIGER; (v) firefighters; (vi) the Colombian Council for Sustainable Construction; (vii) the Urban Development Institute (IDU); and (viii) the Bogotá Aqueduct Company (EEAB).

62. To ensure successful execution of the six prioritized actions, three national mechanisms and funding sources were identified: (i) The FNGRD²⁵ given that the proposed actions are aimed at reducing the vulnerability of San Carlos neighborhood to climate change, this fund supports the financing of projects related to Disaster Risk Management at both national and territorial levels. The alignment of objectives paves the way to the application of these resources following the principles of concurrence and positive subsidiarity; (ii) the Adaptation Fund, which could structure and implement comprehensive projects for risk reduction and adaptation to climate change, within the framework of the National Disaster Risk Management System and in coordination with the respective sectors; and finally, (iii) the Presidential Agency for International Cooperation of Colombia (APC), tasked with managing, guiding, and technically coordinating public, private, technical, and financial non-reimbursable international cooperation received and granted by the country.

63. During the analysis, some constraints or challenges were identified, particularly concerning five specific areas related to the implementation of the proposed Actions. In response, potential solutions were recommended to the district:

(i) Planning, Structuring, and Execution Scope: The proposed actions are currently nested under the District Plan for Economic, Social, Environmental, and Public Works Development 2020-2024. However, with the upcoming

mayoral elections, this plan is slated for modifications. Additionally, the PAC's relevance is anticipated to wane with the introduction of a district-level Conpes that the SDA has been formulating. Given this impending shift, it's essential for the proposed actions to align with national-level strategic instruments. These include the Conpes 3700 of 2011, the Nationally Determined Contribution (NDC), the National Plan for Climate Change Adaptation (PNACC), and the Colombian Low-Carbon Development Strategy (ECDBC). Furthermore, they should be congruent with the Comprehensive Regional Climate Change Plan for Bogotá Cundinamarca (PRICC). Alongside these national strategies, it's crucial to anchor these initiatives to the 'Reverdecer Bogotá' Land Use Plan, which remains valid from 2022 through 2035.

(ii) Financial scope: Considering the significant investment and several work from the multi-faceted work that the proposed actions entail for the San Carlos neighborhood, it is imperative to adopt a well-thought-out plan. The aim is to preemptively address potential disruptions, including mobility collapse, heightened dissatisfaction among residents, work delays, and possible citizen protests. In view of this, a phased implementation of the six prioritized actions is suggested. This structured approach not only ensures that the proposed actions are coordinated in a holistic manner, but it also optimizes institutional efforts and resource management. Striking a balance between the climate action goals and maintaining the social fabric and functionality of the neighborhood can pave the way for a more comprehensive and sustainable outcome.

(iii) Institutional Scope: The presence of key actors identified actors and various works developed by district entities in the San Carlos neighborhood, each with their own agendas and interests, may result in causing a lack of institutional convergence. Therefore, it is crucial to establish robust channels for inter-institutional collaboration and to define clear institutional roles. It is important to consider the significant roles of the SDA, given the unique characteristics of the neighborhood, as well as the SDM, given its city-wide leadership of the Low-Carbon and Vital Neighborhood (LCVN) strategy. Furthermore, there are other actors, such as the that the José Celestino Mutis Botanical Garden, that possess the requisite competencies and resources to develop specific elements of the initiatives. These actors need to be identified and appropriately integrated into the institutional framework.

²⁵National Disaster Risk Management Fund

(iv) Social scope: Despite concerted efforts by the district, some residents expressed skepticism about potential urban interventions. Therefore, it is necessary to socialize the benefits of the strategy to neighborhood residents and hold negotiation meetings, articulate it with the 'Plan de Dinamización' led by SDM, to reach contingency agreements, minimizing any potential impact, with a commitment to compensate for the foreseeable but unintentional impact.

(v) Technical Scope: Currently, detailed engineering is not available as the consultancy's scope was constrained to providing only basic conceptual designs as an initial input for solutions. Therefore, it becomes necessary to develop proposals with a greater level of detail. This development should be undertaken at both the feasibility and detailed engineering stages, encompassing elements such as scaled floor plans, sections, elevations, views, detailed specifications, technical coordination, and budgeting.



6. NEXT STEPS

64. Bogota has the potential to transform from a big megalopolis into a network of low-carbon and climate-resilient neighborhoods. However, to achieve this, it is crucial to address sustainability threats such as limited budget allocation, public legitimacy, institutional arrangements, and community engagement. The city needs to foster an inter-institutional consensus around the benefits and opportunities of the LCVN Strategy. By doing so, it can facilitate urban transformations, and scale-up to establish a broad policy and a public platform that consolidates the 15-minute city paradigm while attending major goals established in Bogota's Climate Action Plan (PAC). This approach not only aims to reduce carbon footprint but also seeks to maximize environmental, social, and economic impacts, creating a new territorial model.

65. From an institutional perspective, the incorporation of three key elements into the regulations of the 'Institutional Convergence Scheme' is essential: (i) enabling the District Mobility Secretariat (SDM) to participate in specific strategies; (ii) integrating the LCVN Strategy into the land use strategy, which should subsequently be incorporated into the District Development Plan; and (iii) ensuring that the Convergence Scheme assigns the necessary departments to implement the various intervention strategies of the POT.

66. Intersectoral coordination with secretaries such as the District Habitat Secretariat (SDH) and District Woman's Secretariat (SDMujer), is vital for two main reasons: (i) to establish an intersectoral agreement that re-defines technical variables and criteria for selecting future intervention zones in the Tactical Urbanism phase, ensuring a multi-dimensional approach beyond the mobility sector; and (ii) to coordinate the LCVN Strategy with other ongoing processes or programs, such as execution of the District Care System through Care Blocks, to facilitate resource convergence.

67. Under the leadership of the District Planning Secretariat (SDP) it would be beneficial to consider the inclusion of pilot projects within the upcoming UPL regulation. This would enable each pilot to become a 'Comprehensive Project' of local interest, aligning them with other strategies contained in the POT to take advantage of the potential to apply land management instruments and leverage financing.

68. The success of the three pilot projects analyzed in this study hinges on the involvement of specific key players. In the case of San Felipe, the District Mobility Secre-

tariat (SDM), District Culture Secretariat (SDCRD), and District Economic Development Secretariat (SDE) should be engaged. For San Carlos, the District Environment Secretariat (SDA), Road Maintenance Unit (UMV), and Bogota Aqueduct Company should be included. Finally, the District Woman's Secretariat (SDMujer) and Heritage Institute (IDPC) should be engaged in the Historic Center.

69. In terms of urban design, the first step involves implementing the Urban Design Guidelines in the rest of the LCVN strategy and other areas of the city, ensuring the inclusion of 'Ecosystemic urbanism' and 'Superblocks' principles. Biodiversity, Nature-based solutions (NbS), urban re-greening, resilient infrastructure, and Sustainable Urban Drainage Systems (SUDS), should be incorporated in the upcoming political agenda, such as the District Development Plan.

70. Simultaneously, it is necessary to establish and implement a robust monitoring and evaluation system. This involves making the dashboard available on the SDM website, to demonstrate the positive effects of the strategy, thereby increasing awareness about the need to replicate and escalate to Urban transformations. The measurement of Tactical urbanism impacts could facilitate future resource allocation for civil works.

71. For Tactical urbanism interventions approved for civil works, the stages determined by the Superblocks' master plan for each neighborhood should be considered to ensure financial feasibility. Additionally, it is crucial to prepare, tender, and execute the enabling infrastructure investments.

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ANNEXES

Annex 1. Overall program cost LCVN

a) Cost Summary (33 VN)

33 Vital Neighborhoods (VN)	
Average area for VN (Ha)	67,6
Estimated cost/Ha (USD)	\$ 902.282
Cost for 33 VN (USD)	\$ 2.012.810.686

(b) Cost estimate by stages (three prioritized neighborhoods)

	San Felipe (VN)	San Carlos (VN)	Historic Center (VN)
Tactical Urbanism (Pilot)	Already implemented by the city	To be determined and implemented by the city	\$ 313.031
Stage 1	\$ 11.195.673	\$ 13.709.662	-
Stage 2	\$ 12.791.286	\$ 27.470.835	-
Stage 3	\$ 8.401.200	\$ 12.167.424	-
Stage 4	\$ 6.373.209	-	-
Stage 5*	\$ 3.809.540	-	-
Total costs (tactical urbanism) to be financed by the VN program (USD)	Already implemented by the city	To be determined and implemented by the city	\$ 313.031
Total costs (infrastructure project) to be financed by the VN program (USD)	\$ 38.761.368	\$ 53.347.920	To be determined by future design

Note a: Exchange rate of US\$1 to COP\$4,430.

Note b: To estimate the total cost of the 33 Vital Neighborhoods program, several assumptions are made: (i) The total intervention area for the first five pilot projects is averaged, and the area of the other 28 Vital Neighborhoods will be similar to the areas studied; (ii) it is assumed that the cost per hectare in all Vital Neighborhoods will be an average of San Felipe and San Carlos projects; (iii) San Felipe and San Carlos were approached as infrastructure projects while Historic Center is developed as a tactical urbanism pilot; (iv) The preliminary costs for the execution of the conceptual design were based on the estimated 2022 reference values of the IDU corresponding to the right-of-ways for the city of Bogotá; (v) The adjustment of 16 percent on the Minimum Legal Wage in force was taken into account for the development of the exercise.

Note c: General lengths and areas were quantified for the main chapters of the project. The unit prices were taken from the IDU price list.

Annex 2. List of indicators and main sources of information

Id	Level*	Policy aspect	Indicator	Main source of information
Category: Sustainable mobility				
M1	1	Road safety	Road accidents	Historical data on traffic accidents in Bogota D.C. 2015-2022. • https://datos.movilidadbogota.gov.co/datasets/movilidadbogota::historico-siniestros-bogota%C3%A1-d-c-explore?location=4.746516%2C-74.138215%2C11.00 • https://datos.movilidadbogota.gov.co/datasets/movilidadbogota::accidente-1/explore?location=4.408655%2C-74.187553%2C10.26
M2	2	Road safety	Speed	Bitcarrier: Intelligent information and monitoring system of traffic flow in real time - It works from WiFi and Bluetooth wireless sensors emitted by mobile devices) • https://www.simur.gov.co/indicadores/velocidades-bitcarrier
M3	1	Transport mode	Transport mode	Households' mobility surveys 2019 (Number and distances of trips per mode) • https://www.simur.gov.co/encuestas-de-movilidad#collapse=accordion-232-1 Interception Surveys in San Felipe held by SDM
M4	2	Transport mode	Type of trips	Households' mobility surveys 2019 (Number and distances of trips per mode) • https://www.simur.gov.co/encuestas-de-movilidad#collapse=accordion-232-1 Interception Surveys in San Felipe held by SDM
M5	1	Performance	Buses accessibility	Location of BRT station and conventional bus stops de Transmilenio. 2022 Residential land use map: 2022 • https://datosabiertos.bogota.gov.co/dataset/mapa-de-referencia
Category: Environmental quality				
A1	1	Air quality	Changes in PM concentration	Information from the PM microsensors network, and from the air quality network stations • http://201.245.192.252:81/Report/stationreport
A2	2	Air quality	Air quality perceived	Citizenship perception survey in San Felipe. Data gathered by SDM.
A3	3	Air quality	CO2 emissions	Vehicle Kilometers Traveled: Households' mobility surveys (2019) Emission Factors: Bogota's Mobility Master Plan (2020)
A4	2	Noise	Perception of noise	Petitions, complaints, and claims related to noise pollution. District Environment Secretariat - SDA (2022) GIS service of the Center for Information and Environmental Modeling of Bogotá (CIMAB).
A5	3	Noise	Noise pressure level	Strategic noise maps. Geographic viewer of the District Environment Secretariat - SDA
A6	3	Visual pollution	Number of illegal advertisement installed	Minor Records of Visual Outdoor Advertising. Geographic viewer of the District Environment Secretariat - SDA
Category: Public space				
E1	1	Quality	Urban trees and woodland	Urban trees census (2022) • https://datosabiertos.bogota.gov.co/dataset/censo-arbolado-urbano
E2	2	Quality	Green areas	Sentinel-2 products (for the calculation of the normalized difference vegetation index - NDVI) • https://dataspace.copernicus.eu/
E3	3	Quality	Public space satisfaction	Citizenship perception survey in San Felipe. Data gathered by SDM.
E4	3	Quality	Pedestrian space	Public Space inventory - Land Use Master Plan - POT (2022) • https://services8.arcgis.com/2gedZBw4OrdjUJLOA/arcgis/rest/services
E5	1	Use of space	Conflicts for public space use	National Registry of Corrective Measures - RNMC (2022). Cadastral sector information. 'Occupying public space in violation of current regulations' • https://oai.ee.scj.gov.co/agc/rest/services/Tematicos_Pub/CifrasSCJ/FeatureServer •LB: https://datosabiertos.bogota.gov.co/dataset/medida-correctiva-bogota-d-c
E6	3	Use of space	Number of people entering commercial locals	Field data gathering by SDM: Number of entries to local businesses on Wednesdays and Saturdays in San Felipe (2022)
E7	1	Supply and access	Public space per capita	Public Space inventory - Land Use Master Plan - POT (2022) Population: Information by cadastral sector and block - DANE (2018) Population growth projection 2018-2024. Information by Zonal Planning Units (UPZ) - DANE (2018) • https://services8.arcgis.com/2gedZBw4OrdjUJLOA/arcgis/rest/services • https://sdpbogota.maps.arcgis.com/sharing/rest/content/items/65f22c5ab6254804ac7b1a8d22afe5ef/data • https://sdpbogota.maps.arcgis.com/sharing/rest/content/items/94e585d7ab404c3aad5b9e08e7ea4689/data

Id	Level*	Policy aspect	Indicator	Main source of information
Category: Community				
C1	1	Security	Number of thefts and robberies	Criminal and Misdemeanor Statistical Information System (SIEDCO) - National Police Department.
C2	2	Security	Violence incidence	High impact crimes: 2018-2022. Cadastral sector information Incidents: 2018-2022. Cadastral sector information • https://oaiee.scj.gov.co/agc/rest/services/Tematicos_Pub/CifrasSCJ/FeatureServer
C3	2	Security	Violent crime rate	Population growth projection 2018-2024. Information by Zonal Planning Units (UPZ) - DANE (2018) • https://oaiee.scj.gov.co/agc/rest/services/Tematicos_Pub/CifrasSCJ/FeatureServer • https://sdpbogota.maps.arcgis.com/sharing/rest/content/items/65f22c5ab6254804ac7b1a8d22afe5ef/data • https://sdpbogota.maps.arcgis.com/sharing/rest/content/items/94e585d7ab404c3aad5b9e08e7ea4689/data
C4	3	Security	Perception of security	Citizenship perception survey in San Felipe. Data gathered by SDM.
C5	1	Access to services and resources	Access to health facilities	Reference map of Bogota. Uses by property and public space (2011 - 2022) • https://datosabiertos.bogota.gov.co/dataset/mapa-de-referencia
C6	2	Access to services and resources	Access to cultural facilities	Cultural facilities from GIS open data (2022) • https://datosabiertos.bogota.gov.co/dataset/mapa-de-referencia

*Conventions:

Level 1: Ready to report using open data

Level 2: Require secondary information processing

Level 3: Require field gathering of primary information

Annex 3. Cost estimation for the six adaptation and mitigation actions in San Carlos Neighborhood

Action	Cost (\$US)	Share (%)
1. Re-naturalization of effective public space and sidewalks	\$ 146.086	3,55%
2. Pedestrianization of roads to increase effective public space	\$ 1.785.255	43,43%
3. Increase Tree Cover	\$ 397.956	9,68%
4. Water care	\$ 56.148	1,37%
5. Mobility and sustainable transport	\$ 446.471	10,86%
6. Green infrastructure - green roofs	\$ 1.278.325	31,10%
Total	\$ 4.110.242	100%

Source: Ecosimple, 2023

Note: Exchange rate of US\$1 to COP\$4,700

Annex 4. Activities developed during the World Bank technical assistance

Consultancy firm	Scope	Description of Quality	Date
JFP Asociados	Policy and regulatory framework recommendations and Management model for LCVN	Site visit at San Felipe Vital neighborhood and interview to local leaders	9/2/2022
JFP Asociados	Policy and regulatory framework recommendations and Management model for LCVN	Work session at the District Mobility Secretariat Office	10/9/2022
JFP Asociados	Policy and regulatory framework recommendations and Management model for LCVN	Institutional Workshop with Secretaries and district agencies	10/20/2022
Fundación Ecología Urbana	Urban design guidelines for LCVN	International Site visit to LCVN Historic Center with City officials	10/29/2022
Fundación Ecología Urbana	Urban design guidelines for LCVN	International Site visit to LCVN San Felipe and San Carlos with City officials	10/30/2022
Fundación Ecología Urbana	Urban design guidelines for LCVN	Preliminary meeting with Secretaries and district agencies (SDM, SDA, SDP, IDU)	10/31/2022
Fundación Ecología Urbana	Urban design guidelines for LCVN	Co-creative workshop with stakeholders and City officials	10/31/2022
JFP Asociados and Fundación Ecología Urbana	Urban design guidelines and Policy recommendations for LCVN	Bogota Mayors Office (Claudia Lope) LCVN Strategy presentation	11/1/2022
Despacio	Policy recommendations for Bogota's Mobility Plan and Policy and regulatory framework recommendations and Management model for LCVN	Work meeting session and feedback to PMSS	11/10/2022
Fundación Ecología Urbana and WB	Technical assistance to VN	Work session: Vital Neighborhoods & Care blocks (SDH)(SDH, SDM, SDMujer, SDA)	12/6/2022
Despacio	Policy recommendations for Bogota's Mobility Plan	Final presentation and next steps	12/13/2022
Ecosimple	Technical assistance to VN	Work meeting: Vital Neighborhoods & Bogota's	1/20/2023
WB	Technical assistance to VN	Climate Action Plan (Resilient infrastructure)(SDA)	1/23/2023
Ecosimple	Portfolio of adaptation and mitigation actions for climate change	Work meeting: Vital Neighborhoods & Care blocks (SDH, SDM)	1/31/2023
Hill Consulting	Environmental impact assessment for LCVN	Site visit with local stakeholders: San Carlos Neighborhood.	1/31/2023
Ecosimple	Portfolio of adaptation and mitigation actions for climate change	Focus group with district secretaries: Priorization of baseline indicators	2/10/2023
Hill Consulting	Environmental impact assessment for LCVN	Focus group with Stakeholders	2/15/2023
Hill Consulting	Environmental impact assessment for LCVN	Asicronic complement, prioritization exercise Mentimeter (SDA, SDM, SDDE)	2/16/2023
Hill Consulting	Environmental impact assessment for LCVN	Presentation partial results Environmental impact assessment for San Felipe (SDA)	3/3/2023
Taller Architects	Pre-feasibility studies kr22 in San Felipe	Site visit El Porvernir Care Block in Bosa (SDMujer, SDM)	3/15/2023
Hill Consulting	Environmental impact assessment for LCVN	Pilot meeting with Stakeholders, local leaders, district agencies	3/17/2023
Ecosimple	Portfolio of adaptation and mitigation actions for climate change	Work session: Indicators sheets review (WRI, SDA) Intersectoral workshop: Multicriteria matrix exercise for San Carlos VN	3/29/2023
Taller Architects	Pre-feasibility studies kr22 in San Felipe	Work session: technical and legal procedure review (EEAB)	3/29/2023
WB	Urban design guidelines for LCVN	Articulation session Vital neighborhoods and Care Blocks (SDM, SDMujer, IDPC, SDH)	4/11/2023
Taller Architects	Pre-feasibility studies kr22 in San Felipe	Work session: technical review - prefeasibility stage (studies and designs)(IDU, SDM)	4/26/2023
WB	Urban design guidelines for LCVN	Work session San Felipe, San Carlos and Historic center (SDM, SDMujer)	5/16/2023
Ecosimple	Portfolio of adaptation and mitigation actions for climate change	Work session: LCVN & Climate Action Plan articulation session (SDA)	5/17/2023

Note: The technical assistance provided by the World Bank to the city of Bogota, counted on with active participation from the Mayors Office, to the District Mobility Secretariat (SDM), District Mobility Secretariat (SDA), and District Planning Secretariat (SDP). Several secretaries and agencies such as the District Woman's Secretariat (SDMujer), District Habitat Secretariat (SDH), District Economic Development Secretariat (SDEE), District Economic Development Secretariat, and the Urban Development institute (IDU), Bogota's Botanical Garden, District Heritage and Culture Institute (IDPC), Road Maintenance Unit (UMV), Administrative Department for Public Space (DADEP), and Aqueduct and Sewerage Company of Bogotá (EEAB), were deeply involved along the process. Co-creative workshops, site visits, community sessions with different stakeholders, international experts, consultants, local leaders and community took also place.

